

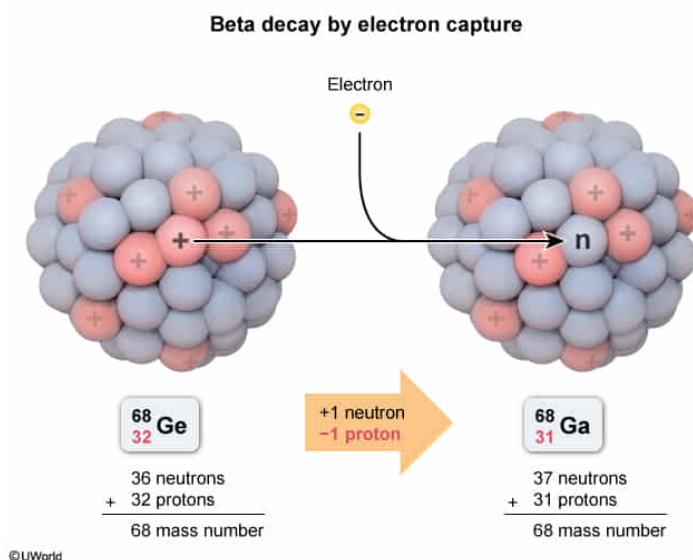
The radioisotope gallium-68 is used in nuclear medicine as a traceable radioactive emitter for diagnostic imaging procedures. The ^{68}Ga isotope is generated from ^{68}Ge by:

- ☐ A. adding a proton to the nucleus. [5%]
- ✓ ☒ B. converting a proton in the nucleus to a neutron by electron capture. [64%]
- ☐ C. converting a neutron in the nucleus to a proton by β^- decay. [27%]
- ☐ D. removing three electrons from the atom. [2%]

Correct

64% answered correctly

Explanation:



Radioactive **beta decay** can occur in three forms: β^- decay (electron emission), β^+ decay (positron emission), and electron capture. In all three forms of beta decay, the **mass number** remains unchanged whereas the **atomic number** either increases (β^- decay) or decreases (β^+ decay and electron capture) by 1. As the atomic number changes, the identity of the element changes accordingly. During electron capture, a nuclear **proton** captures an inner electron near the nucleus and then converts to a **neutron**. This decreases the atomic number by

1 but leaves the mass number the same.

In the conversion of ^{68}Ge into ^{68}Ga , the identity of the element changes; therefore, the number of protons (atomic number) must decrease by 1. Because the mass number of 68 is unchanged, the loss of 1 proton must then be accompanied by the gain of 1 neutron. Therefore, ^{68}Ga is generated from ^{68}Ge by converting a proton to a neutron (as occurs during electron capture).

(Choice A) Adding a proton to the ^{68}Ge nucleus would increase the mass number by 1 and change its identity to arsenic to give ^{69}As instead of ^{68}Ga .

(Choice C) Converting a neutron to a proton in the ^{68}Ge nucleus by β^- decay leaves the mass number unchanged, but the atomic number would still increase by 1 and change the identity of the isotope to ^{68}As .

(Choice D) Removing three electrons from ^{68}Ge would only produce a $^{68}\text{Ge}^{3+}$ ion. It would not alter the nucleus (number of protons) and would not change the identity of the element from ^{68}Ge to ^{68}Ga .

Educational objective:

Nuclear conversions may change the atomic number (number of protons) or the mass number (combined number of protons and neutrons). In beta decay, a proton can convert to a neutron (or vice versa), leaving the mass number unchanged.

Time spent:0 sec

Subject:
General Chemistry

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System:
Atomic Theory and Chemical Composition